

MERT CIHAN BAYIR

Software Engineer — Artificial Intelligence — Machine Learning — Mobile AI

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SUMMARY

Software Engineer specialized in Artificial Intelligence, Machine Learning, Computer Vision, and Kotlin-based mobile applications. Experienced in end-to-end AI pipelines including LiDAR data annotation, feature engineering, model training, mobile deployment (TFLite), RAG-based chatbot development, workflow automation, and containerized systems.

EDUCATION

Sakarya University – Software Engineering

2021 – 2025

- Bachelor's Degree in Software Engineering.

EXPERIENCE

VEXPO – AI Engineer

2026 -

- Led the end-to-end design and development of a RAG (Retrieval-Augmented Generation) based chatbot system.
- Implemented document processing pipelines, vector database integration, and retrieval mechanisms.
- Optimized chatbot response quality through LLM integration and prompt engineering techniques.
- Developed an intelligent assistant capable of providing context-aware and accurate responses to user queries.

Ford Otosan – Data Annotator

2023 – 2025

- Annotated large-scale LiDAR point cloud datasets for autonomous driving perception systems.
- Labeled 3D objects such as vehicles, pedestrians, and road elements to support detection and tracking models.
- Performed annotation validation and quality control to ensure dataset consistency and accuracy.

INTERNSHIPS

Ford Otosan – Artificial Intelligence Intern

2024

- Developed a deep learning-based assistant model to accelerate data annotation workflows.
- Improved labeling efficiency by reducing manual annotation time.

MKU Technology – Artificial Intelligence Intern

2024

- Worked in the AI team of an early-stage gender detection project.
- Contributed to data preprocessing, feature extraction, and model evaluation pipelines.

PROGRAMS

Google Artificial Intelligence and Technology Academy

2025

- Completed Google-certified Artificial Intelligence and Machine Learning courses.
- Developed AI-driven projects during hackathon and bootcamp programs.
- Built a React-based platform generating fully AI-driven personalized daily learning plans.
- Designed an educational AI system that guides users to discover answers instead of providing direct responses.
- Completed entrepreneurship and innovation-focused training.

PROJECTS

Turkish Coffee Fortune Telling AI Application

Graduation Project

- Created and manually annotated a custom dataset of 8,049 images.
- Trained a YOLOv11n classification model achieving 99.5% accuracy.
- Converted the model to TFLite and integrated it into a Kotlin-based Android application.
- Generated personalized fortune interpretations based on detected cup type.

Early Gender Detection

Machine Learning

- Converted raw audio data into structured datasets using feature extraction and normalization techniques.
- Developed a Random Forest model achieving 96.4% accuracy.

EEG-Based Yes / No Detection

Machine Learning

- Collected EEG data using a 14-channel Emotiv EEG device.
- Extracted signal-based features and trained a Random Forest model achieving approximately 90% accuracy.
- Successfully detected real-time cognitive yes/no responses.

SKILLS

Programming: Python, Java, Kotlin, JavaScript, TypeScript

Mobile Development: Android, Kotlin, TFLite Integration

Machine Learning: Random Forest, SVM, Feature Engineering, Signal Processing

Deep Learning: YOLO, CNNs

NLP / LLM: RAG, LangChain, Vector Databases, Prompt Engineering

Automation: n8n, Workflow Automation

DevOps: Docker, Kubernetes, Git, GitHub

Data: LiDAR Annotation, Data Collection, Dataset Balancing